

How MOBILE PHONES HAVE EVOLVED And Lead To NEWER INNOVATIONS

It's a pretty exciting time for mobile architecture since the world is currently struggling with the gradual decline of Moore's Law. In his keynote address at the VLSI Design Conference 2022, Ken Wiseman (VP - Technology, Qualcomm) talked about mobile architecture, the existing challenges, and how the mobile architecture has led to newer innovations



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Cellphones have influenced the world more than any other electronic device. Just ask yourself: How many times do you check your phone in a day? In the beginning, mobile phones were used for making calls only while moving around. The ability to text other mobile phones was introduced shortly thereafter. Nowadays, you can do practically anything on a mobile device.

"In 1992, I actually worked on my first phone ever in Japan. It was a sys-

tem called PHS (Personal Handy-phone System), which was a microcellular technology. It was just a basic version only for calls, and there was no data," says Ken Wiseman, VP - Technology, Qualcomm. Today, we have true mobility and performance in our phones.

You have now got a computer in your hand—there is multimedia, 4K Ultra HD video player, recorder, navigation, broadband, and much more. You can download files, interact on Facebook or other types of platforms, and share experiences. There is interconnectivity anywhere, whether you have Wi-Fi or cellular—there is always a connected device in your hand today. At first, battery life was just about a few hours. Now, we can keep our devices powered for a very long time. The architecture and complexity have also evolved. So how did this happen?

How did the capabilities of these handheld devices evolve so quickly?

Moore's Law has been a big piece of that enablement. Today, Moore's Law has definitely slowed down. In his paper titled *What's Next? (The end of Moore's Law)*, R. Stanley Williams, research scientist for HP Labs writes, "The end of Moore's Law may be the best thing that has happened in computing since the beginning of Moore's Law. Confronting the end of an epoch should enable a new era of creativity..."

"In 1999, I was working on the PHS handset," says Wiseman. "My company



On December 3, 1992, Neil Papworth—a test engineer for Sema Group—sent the first text message to a mobile phone. It just said, "Merry Christmas"